

Proposition 1. Let I be a nonempty countable set and for each $i \in I$ let A_i be a set. The cardinality of the Cartesian product is the product of the cardinalities of the sets A_i , i.e.,

$$|\prod_{i \in I} A_i| = \prod_{i \in I} |A_i|,$$

(where if some A_i is an infinite set or if I is infinite and an infinite number of A_i 's have cardinality ≥ 2 , both sides of this equality are infinity). In particular,

$$|A_1 \times A_2 \times \cdots \times A_n| = |A_1| \times |A_2| \times \cdots \times |A_n|.$$

Proof: In order to count the number of choice functions note that each $i \in I$ may be mapped to any of the $|A_i|$ elements of A_i and for $i \neq j$ the values of choice functions at i and j may be chosen completely independently. Thus the number of choice functions is the product of the cardinalities of the A_i 's, as claimed.

For Cartesian products of finitely many sets, $A_1 \times A_2 \times \cdots \times A_n$, one can see this easily from the n -tuple representation: the elements of $A_1 \times A_2 \times \cdots \times A_n$ are n -tuples $(a_1, a_2, \dots, a_n)$ and each a_i may be chosen as any of the $|A_i|$ elements of A_i . Since these choices are made independently for $i \neq j$, there are $|A_1| \cdot |A_2| \cdots |A_n|$ elements in the Cartesian product.

EXERCISE

1. Let I and J be any two indexing sets and let A be an arbitrary set. For any function $\varphi : J \rightarrow I$ define

$$\varphi^* : \prod_{i \in I} A \rightarrow \prod_{j \in J} A \quad \text{by} \quad \varphi^*(f) = f \circ \varphi \quad \text{for all choice functions } f \in \prod_{i \in I} A.$$

- (a) Let $I = \{1, 2\}$, let $J = \{1, 2, 3\}$ and let $\varphi : J \rightarrow I$ be defined by $\varphi(1) = 2$, $\varphi(2) = 2$ and $\varphi(3) = 1$. Describe explicitly how a 3-tuple in $A \times A \times A$ maps to an ordered pair in $A \times A$ under this φ^* .
- (b) Let $I = J = \{1, 2, \dots, n\}$ and assume φ is a permutation of I . Describe in terms of n -tuples in $A \times A \times \cdots \times A$ the function φ^* .

2. PARTIALLY ORDERED SETS AND ZORN'S LEMMA

We shall have occasion to use Zorn's Lemma as a form of "infinite induction" in a few places in the text where it is desirable to know the existence of some set which is *maximal* with respect to certain specified properties. For example, Zorn's Lemma is used to show that every vector space has a basis. In this situation a basis of a vector space V is a subset of V which is maximal as a set consisting of linearly independent vectors (the maximality ensures that these vectors span V). For finite dimensional spaces this can be proved by induction; however, for spaces of arbitrary dimension Zorn's Lemma is needed to establish this. By having results which hold in full generality the theory often becomes a little neater in places, although the main results of the text do not require its use.

A specific instance in the text where a maximal object which helps to simplify matters is constructed by Zorn's Lemma is the algebraic closure of a field. An algebraic closure of a field F is an extension of F which is maximal among any collection of algebraic extensions. Such a field contains (up to isomorphism) all elements which are algebraic over F , hence all manipulations involving such algebraic elements can be effected in this one larger field. In any particular situation the use of an algebraic closure can be avoided by adjoining the algebraic elements involved to the base field F , however this becomes tedious (and often obscures matters) in complicated proofs. For the specific fields appearing as examples in this text the use of Zorn's Lemma to construct an algebraic closure can be avoided (for example, the construction of an algebraic closure of any subfield of the complex numbers or of any finite field does not require it).

The first example of the use of Zorn's Lemma appears in the proof of Proposition 11 in Section 7.4.

In order to state Zorn's Lemma we need some terminology.

Definition. A *partial order* on a nonempty set A is a relation $\leq$ on A satisfying

- (1) $x \leq x$ for all $x \in A$ (reflexive),
- (2) if $x \leq y$ and $y \leq x$ then $x = y$ for all $x, y \in A$ (antisymmetric),
- (3) if $x \leq y$ and $y \leq z$ then $x \leq z$ for all $x, y, z \in A$ (transitive).

We shall usually say that A is a partially ordered set under the ordering $\leq$ or that A is partially ordered by $\leq$.

Definition. Let the nonempty set A be partially ordered by $\leq$.

- (1) A subset B of A is called a *chain* if for all $x, y \in B$, either $x \leq y$ or $y \leq x$.
- (2) An *upper bound* for a subset B of A is an element $u \in A$ such that $b \leq u$, for all $b \in B$.
- (3) A *maximal element* of A is an element $m \in A$ such that if $m \leq x$ for any $x \in A$, then $m = x$.

In the literature a chain is also called a *tower* or called a *totally ordered* or *linearly ordered* or *simply ordered* subset.

Some examples below highlight the distinction between upper bounds and maximal elements. Also note that if m is a *maximal* element of A , it is not necessarily the case that $x \leq m$ for all $x \in A$ (i.e., m is not necessarily a *maximum* element).

Examples

- (1) Let A be the power set (i.e., set of all subsets) of some set X and $\leq$ be set containment: $\subseteq$. Notice that this is only a *partial* ordering since some subsets of X may not be comparable, e.g. singletons: if $x \neq y$ then $\{x\} \not\subseteq \{y\}$ and $\{y\} \not\subseteq \{x\}$. In this situation an example of a chain is a collection of subsets of X such as

$$X_1 \subseteq X_2 \subseteq X_3 \subseteq \cdots.$$

Any subset B of A has an upper bound, b , namely,

$$b = \bigcup_{x \in B} x.$$

This partially ordered set A has a (unique) maximal element, X .

In many instances the set A consists of some (but not necessarily all) subsets of a set X (i.e., A is a subset of the power set of X) and with the ordering on A again being inclusion. The existence of upper bounds and maximal elements depends on the nature of A .

- (2) Let A be the collection of all *proper* subsets of $\mathbb{Z}^+$ ordered under $\subseteq$. In this situation, chains need not have maximal elements, e.g. the chain

$$\{1\} \subseteq \{1, 2\} \subseteq \{1, 2, 3\} \subseteq \cdots$$

does not have an upper bound. The set A does have maximal elements: for example $\mathbb{Z}^+ - \{n\}$ is a maximal element of A for any $n \in \mathbb{Z}^+$.

- (3) Let $A = \mathbb{R}$ under the usual $\leq$ relation. In this example every subset of A is a chain (including A itself). The notion of a subset of A having an upper bound is the same as the usual notion of a subset of $\mathbb{R}$ being bounded above by some real number (so some sets, such as intervals of finite length, have upper bounds and others, such as the set of positive reals, do not). The set A does not have a maximal element.

Zorn's Lemma If A is a nonempty partially ordered set in which every chain has an upper bound then A has a maximal element.

It is a nontrivial result that *Zorn's Lemma is independent of the usual (Zermelo–Fraenkel) axioms of set theory*¹ in the sense that if the axioms of set theory are consistent,² then so are these axioms together with Zorn's Lemma; and if the axioms of set theory are consistent, then so are these axioms together with the *negation* of Zorn's Lemma. The use of the term “lemma” in Zorn's Lemma is historical.

For the sake of completeness (and to relate Zorn's Lemma to formulations found in other courses) we include two other equivalent formulations of Zorn's Lemma.

The Axiom of Choice The Cartesian product of any nonempty collection of nonempty sets is nonempty. In other words, if I is any nonempty (indexing) set and A_i is a nonempty set for all $i \in I$, then there exists a choice function from I to $\cup_{i \in I} A_i$.

Definition. Let A be a nonempty set. A *well ordering* on A is a total ordering on A such that every nonempty subset of A has a minimum (or smallest) element, i.e., for each nonempty $B \subseteq A$ there is some $s \in B$ such that $s \leq b$, for all $b \in B$.

The Well Ordering Principle Every nonempty set A has a well ordering.

Theorem 2. Assuming the usual (Zermelo–Fraenkel) axioms of set theory, the following are equivalent:

- (1) Zorn's Lemma
- (2) the Axiom of Choice
- (3) the Well Ordering Principle.

Proof: This follows from elementary set theory. We refer the reader to *Real and Abstract Analysis* by Hewitt and Stromberg, Springer-Verlag, 1965, Section 3 for these equivalences and some others.

¹See P.J. Cohen's papers in: Proc. Nat. Acad. Sci., 50(1963), and 51(1964).

²This is not known to be the case!

EXERCISES

1. Let A be the collection of all finite subsets of $\mathbb{R}$ ordered by inclusion. Discuss the existence (or nonexistence) of upper bounds, minimal and maximal elements (where minimal elements are defined analogously to maximal elements). Explain why this is not a well ordering.
2. Let A be the collection of all infinite subsets of $\mathbb{R}$ ordered by inclusion. Discuss the existence (or nonexistence) of upper bounds, minimal and maximal elements. Explain why this is not a well ordering.
3. Show that the following partial orderings on the given sets are not well orderings:
 - (a) $\mathbb{R}$ under the usual relation $\leq$.
 - (b) $\mathbb{R}^+$ under the usual relation $\leq$.
 - (c) $\mathbb{R}^+ \cup \{0\}$ under the usual relation $\leq$.
 - (d) $\mathbb{Z}$ under the usual relation $\leq$.
4. Show that $\mathbb{Z}^+$ is well ordered under the usual relation $\leq$.

Category Theory

Category theory provides the language and the mathematical foundations for discussing properties of large classes of mathematical objects such as the class of “all sets” or “all groups” while circumventing problems such as Russell’s Paradox. In this framework one may explore the commonality across classes of concepts and methods used in the study of each class: homomorphisms, isomorphisms, etc., and one may introduce tools for studying relations between classes: functors, equivalence of categories, etc. One may then formulate precise notions of a “natural” transformation and “natural” isomorphism, both within a given class or between two classes. (In the text we described “natural” as being “coordinate free.”) A prototypical example of natural isomorphisms within a class is the isomorphism of an arbitrary finite dimensional vector space with its double dual in Section 11.3. In fact one of the primary motivations for the introduction of categories and functors by S. Eilenberg and S. MacLane in 1945 was to give a precise meaning to the notions of “natural” in cases such as this. Category theory has also played a foundational role for formalizing new concepts such as schemes (cf. Section 15.5) that are fundamental to major areas of contemporary research (e.g., algebraic geometry). Pioneering work of this nature was done by A. Grothendieck, K. Morita and others.

Our treatment of category theory should be viewed more as an introduction to some of the basic language. Since we have not discussed the Zermelo–Fraenkel axioms of set theory or the Gödel–Bernays axioms of classes we make no mention of the foundations of category theory. To remain consistent with the set theory axioms, however, we implicitly assume that there is a *universe* set U which contains all the sets, groups, rings, etc. that one would encounter in “ordinary” mathematics (so that the category of “all sets” implicitly means “all sets in U ”). The reader is referred to books on set theory, logic, or category theory such as *Categories for the Working Mathematician* by S. MacLane, Springer–Verlag, 1971 for further study.

We have organized this appendix so that wherever possible the examples of each new concept use terminology and structures in the order that these appear in the body of the text. For instance, the first example of a functor involves sets and groups, the second example uses rings, etc. In this way the appendix may be read early on in one’s study, and a greater appreciation may be gained through rereading the examples as one becomes conversant with a wider variety of mathematical structures.

1. CATEGORIES AND FUNCTORS

We begin with the basic concept of this appendix.

Definition. A *category* C consists of a class of *objects* and sets of *morphisms* between those objects. For every ordered pair A, B of objects there is a set $\text{Hom}_C(A, B)$ of

morphisms from A to B , and for every ordered triple A, B, C of objects there is a *law of composition* of morphisms, i.e., a map

$$\text{Hom}_{\mathbf{C}}(A, B) \times \text{Hom}_{\mathbf{C}}(B, C) \longrightarrow \text{Hom}_{\mathbf{C}}(A, C)$$

where $(f, g) \mapsto gf$, and gf is called the composition of g with f . The objects and morphism satisfy the following axioms: for objects A, B, C and D

- (i) if $A \neq B$ or $C \neq D$, then $\text{Hom}_{\mathbf{C}}(A, B)$ and $\text{Hom}_{\mathbf{C}}(C, D)$ are disjoint sets,
- (ii) composition of morphisms is associative, i.e., $h(gf) = (hg)f$ for every f in $\text{Hom}_{\mathbf{C}}(A, B)$, g in $\text{Hom}_{\mathbf{C}}(B, C)$ and h in $\text{Hom}_{\mathbf{C}}(C, D)$,
- (iii) each object has an identity morphism, i.e., for every object A there is a morphism $1_A \in \text{Hom}_{\mathbf{C}}(A, A)$ such that $f1_A = f$ for every $f \in \text{Hom}_{\mathbf{C}}(A, B)$ and $1_A g = g$ for every $g \in \text{Hom}_{\mathbf{C}}(B, A)$.

Morphisms are also called *arrows*. It is an exercise to see that the identity morphism for each object is unique (by the same argument that the identity of a group is unique). We shall write $\text{Hom}(A, B)$ for $\text{Hom}_{\mathbf{C}}(A, B)$ when the category is clear from the context.

The terminology we use throughout the text is common to all categories: a morphism from A to B will be denoted by $f : A \rightarrow B$ or $A \xrightarrow{f} B$. The object A is the *domain* of f and B is the *codomain* of f . A morphism from A to A is an endomorphism of A . A morphism $f : A \rightarrow B$ is an isomorphism if there is a morphism $g : B \rightarrow A$ such that $gf = 1_A$ and $fg = 1_B$.

There is a natural notion of a *subcategory* category $\mathbf{C}$ of $\mathbf{D}$, i.e., when every object of $\mathbf{C}$ is also an object in $\mathbf{D}$, and for objects A, B in $\mathbf{C}$ we have the containment $\text{Hom}_{\mathbf{C}}(A, B) \subseteq \text{Hom}_{\mathbf{D}}(A, B)$.

Examples

In each of the following examples we leave the details of the verification of the axioms for a category as exercises.

- (1) **Set** is the category of all sets. For any two sets A and B , $\text{Hom}(A, B)$ is the set of all functions from A to B . Composition of morphisms is the familiar composition of functions: $gf = g \circ f$. The identity in $\text{Hom}(A, A)$ is the map $1_A(a) = a$, for all $a \in A$. This category contains the category of all finite sets as a subcategory.
- (2) **Grp** is the category of all groups, where morphisms are group homomorphisms. Note that the composition of group homomorphisms is again a group homomorphism. A subcategory of **Grp** is **Ab**, the category of all abelian groups. Similarly, **Ring** is the category of all nonzero rings with 1, where morphisms are ring homomorphisms that send 1 to 1. The category **CRing** of all commutative rings with 1 is a subcategory of **Ring**.
- (3) For a fixed ring R , the category $R\text{-mod}$ consists of all left R -modules with morphisms being R -module homomorphisms.
- (4) **Top** is the category whose objects are topological spaces and morphisms are continuous maps between topological spaces (cf. Section 15.2). Note that the identity (set) map from a space to itself is continuous in every topology, so $\text{Hom}(A, A)$ always has an identity.
- (5) Let $\mathbf{0}$ be the empty category, with no objects and no morphisms. Let $\mathbf{1}$ denote the category with one object, A , and one morphism: $\text{Hom}(A, A) = \{1_A\}$. Let $\mathbf{2}$ be the category with two objects, A_1 and A_2 , and only one nonidentity morphism:

$\text{Hom}(A_1, A_2) = \{f\}$ and $\text{Hom}(A_2, A_1) = \emptyset$. Note that the objects A_1 and A_2 and the morphism f are “primitives” in the sense that A_1 and A_2 are not defined to be sets and f is simply an arrow (literally) from A_1 to A_2 ; it is not defined as a set map on the elements of some set. One can continue this way and define N to be the category with N objects $A_1, A_2, \dots, A_N$ with the only nonidentity morphisms being a unique arrow from A_i to A_j for every $j > i$ (so that composition of arrows is uniquely determined).

- (6) If G is a group, form the category $\mathbf{G}$ as follows. The only object is G and $\text{Hom}(G, G) = G$; the composition of two functions f and g is the product gf in the group G . Note that $\text{Hom}(G, G)$ has an identity morphism: the identity of the group G .

Definition. Let $\mathbf{C}$ and $\mathbf{D}$ be categories.

- (1) We say $\mathcal{F}$ is a *covariant functor* from $\mathbf{C}$ to $\mathbf{D}$ if
- (a) for every object A in $\mathbf{C}$, $\mathcal{F}A$ is an object in $\mathbf{D}$, and
 - (b) for every $f \in \text{Hom}_{\mathbf{C}}(A, B)$ we have $\mathcal{F}(f) \in \text{Hom}_{\mathbf{D}}(\mathcal{F}A, \mathcal{F}B)$,
- such that the following axioms are satisfied:
- (i) if gf is a composition of morphisms in $\mathbf{C}$, then $\mathcal{F}(gf) = \mathcal{F}(g)\mathcal{F}(f)$ in $\mathbf{D}$, and
 - (ii) $\mathcal{F}(1_A) = 1_{\mathcal{F}A}$.
- (2) We say $\mathcal{F}$ is a *contravariant functor* from $\mathbf{C}$ to $\mathbf{D}$ if the conditions in (1) hold but property (b) and axiom (i) are replaced by:
- (b') for every $f \in \text{Hom}_{\mathbf{C}}(A, B)$, $\mathcal{F}(f) \in \text{Hom}_{\mathbf{D}}(\mathcal{F}B, \mathcal{F}A)$,
 - (i') if gf is a composition of morphisms in $\mathbf{C}$, then $\mathcal{F}(gf) = \mathcal{F}(f)\mathcal{F}(g)$ in $\mathbf{D}$
- (i.e., contravariant functors reverse the arrows).

Examples

In each of these examples the verification of the axioms for a functor are left as exercises. Additional examples of functors appear in the exercises at the end of this section.

- (1) The identity functor $\mathcal{I}_{\mathbf{C}}$ maps any category $\mathbf{C}$ to itself by sending objects and morphisms to themselves. More generally, if $\mathbf{C}$ is a subcategory of $\mathbf{D}$, the *inclusion functor* maps $\mathbf{C}$ into $\mathbf{D}$ by sending objects and morphisms to themselves.
- (2) Let $\mathcal{F}$ be the functor from $\mathbf{Grp}$ to $\mathbf{Set}$ that maps any group G to the same set G and any group homomorphism φ to the same set map φ . This functor is called the *forgetful functor* since it “removes” or “forgets” the structure of the groups and the homomorphisms between them. Likewise there are forgetful functors from the categories $\mathbf{Ab}$, $\mathbf{R-mod}$, $\mathbf{Top}$, etc., to $\mathbf{Set}$.
- (3) The *abelianizing* functor maps $\mathbf{Grp}$ to $\mathbf{Ab}$ by sending each group G to the abelian group $G^{\text{ab}} = G/G'$, where G' is the commutator subgroup of G (cf. Section 5.4). Each group homomorphism $\varphi : G \rightarrow H$ is mapped to the induced homomorphism on quotient groups:

$$\bar{\varphi} : G^{\text{ab}} \rightarrow H^{\text{ab}} \quad \text{by} \quad \bar{\varphi}(xG') = \varphi(x)H'.$$

The definition of the commutator subgroup ensures that $\bar{\varphi}$ is well defined and the axioms for a functor are satisfied.

- (4) Let R be a ring and let D be a left R -module. For each left R -module N the set $\text{Hom}_R(D, N)$ is an abelian group, and is an R -module if R is commutative (cf. Proposition 2 in Section 10.2). If $\varphi : N_1 \rightarrow N_2$ is an R -module homomorphism, then for every $f \in \text{Hom}_R(D, N_1)$ we have $\varphi \circ f \in \text{Hom}_R(D, N_2)$. Thus

$\varphi' : \text{Hom}_R(D, N_1) \rightarrow \text{Hom}_R(D, N_2)$ by $\varphi'(f) = \varphi \circ f$. This shows the map

$$\mathcal{H}om(D, _) : N \longrightarrow \text{Hom}_R(D, N)$$

$$\mathcal{H}om(D, _) : \varphi \longrightarrow \varphi'$$

is a covariant functor from $R\text{-Mod}$ to $\mathbf{Grp}$. If R is commutative, it maps $R\text{-Mod}$ to itself.

- (5) In the notation of the preceding example, we observe that if $\varphi : N_1 \rightarrow N_2$, then for every $g \in \text{Hom}_R(N_2, D)$ we have $g \circ \varphi \in \text{Hom}_R(N_1, D)$. Thus $\varphi' : \text{Hom}_R(N_2, D) \rightarrow \text{Hom}_R(N_1, D)$ by $\varphi'(g) = g \circ \varphi$. In this case the map

$$\mathcal{H}om(_, D) : N \longrightarrow \text{Hom}_R(N, D)$$

$$\mathcal{H}om(_, D) : \varphi \longrightarrow \varphi'$$

defines a *contravariant* functor.

- (6) When D is a right R -module the map $D \otimes_R _ : N \rightarrow D \otimes_R N$ defines a covariant functor from $R\text{-Mod}$ to $\mathbf{Ab}$ (or to $R\text{-Mod}$ when R is commutative). Here the morphism $\varphi : N_1 \rightarrow N_2$ maps to the morphism $1 \otimes \varphi$.

Likewise when D is a left R -module $_ \otimes_R D : N \rightarrow N \otimes_R D$ defines a covariant functor from the category of right R -modules to $\mathbf{Ab}$ (or to $R\text{-Mod}$ when R is commutative), where the morphism φ maps to the morphism $\varphi \otimes 1$.

- (7) Let K be a field and let $K\text{-fdVec}$ be the category of all finite dimensional vector spaces over K , where morphisms in this category are K -linear transformations. We define the *double dual* functor $\mathcal{D}^2$ from $K\text{-fdVec}$ to itself. Recall from Section 11.3 that the dual space, V^* , of V is defined as $V^* = \text{Hom}_K(V, K)$; the double dual of V is $V^{**} = \text{Hom}_K(V^*, K)$. Then $\mathcal{D}^2$ is defined on objects by mapping a vector space V to its double dual V^{**} . If $\varphi : V \rightarrow W$ is a linear transformation of finite dimensional spaces, then

$$\mathcal{D}^2(\varphi) : V^{**} \rightarrow W^{**} \quad \text{by} \quad \mathcal{D}^2(\varphi)(E_v) = E_{\varphi(v)},$$

where E_v denotes “evaluation at v ” for each $v \in V$. By Theorem 19 in Section 11.3, $E_v \in V^{**}$, and each element of V^{**} is of the form E_v for a unique $v \in V$. Since $\varphi(v) \in W$ we have $E_{\varphi(v)} \in W^{**}$, so $\mathcal{D}^2(\varphi)$ is well defined.

The functor $\mathcal{F}$ from $\mathbf{C}$ to $\mathbf{D}$ is called *faithful* (or is called *full*) if for every pair of objects A and B in $\mathbf{C}$ the map $\mathcal{F} : \text{Hom}(A, B) \rightarrow \text{Hom}(\mathcal{F}A, \mathcal{F}B)$ is injective (or surjective, respectively). Thus, for example, the forgetful functor is faithful but not full.

EXERCISES

- Let N be a group and let $\mathbf{Nor}\text{-}N$ be the collection of all groups that contain N as a normal subgroup. A morphism between objects A and B is any group homomorphism that maps N into N .
 - Prove that $\mathbf{Nor}\text{-}N$ is a category.
 - Show how the projection homomorphism $G \mapsto G/N$ may be used to define a functor from $\mathbf{Nor}\text{-}N$ to $\mathbf{Grp}$.
- Let H be a group. Define a map $\mathcal{H} \times$ from $\mathbf{Grp}$ to itself on objects and morphisms as follows:

$$\mathcal{H} \times : G \rightarrow H \times G, \text{ and}$$

$$\text{if } \varphi : G_1 \rightarrow G_2 \text{ then } \mathcal{H} \times(\varphi) : H \times G_1 \rightarrow H \times G_2 \text{ by } (h, g) \mapsto (h, \varphi(g)).$$

Prove that $\mathcal{H}\times$ is a functor.

3. Show that the map **Ring** to **Grp** by mapping a ring to its group of units (i.e., $R \mapsto R^\times$) defines a functor. Show by explicit examples that this functor is neither faithful nor full.
4. Show that for each $n \geq 1$ the map $\mathcal{GL}_n : R \rightarrow GL_n(R)$ defines a functor from **CRing** to **Grp**. [Define $\mathcal{GL}_n$ on morphisms by applying each ring homomorphism to the entries of a matrix.]
5. Supply the details that show the double dual map described in Example 7 satisfies the axioms of a functor.

2. NATURAL TRANSFORMATIONS AND UNIVERSALS

As mentioned in the introduction to this appendix, one of the motivations for the inception of category theory was to give a precise definition of the notion of “natural” isomorphism. We now do so, and see how some natural maps mentioned in the text are instances of the categorical concept. We likewise give the categorical definition of “universal arrows” and view some occurrences of universal properties in the text in this light.

Definition. Let $\mathbf{C}$ and $\mathbf{D}$ be categories and let $\mathcal{F}, \mathcal{G}$ be covariant functors from $\mathbf{C}$ to $\mathbf{D}$. A *natural transformation* or *morphism of functors* from $\mathcal{F}$ to $\mathcal{G}$ is a map η that assigns to each object A in $\mathbf{C}$ a morphism η_A in $\text{Hom}_{\mathbf{D}}(\mathcal{F}A, \mathcal{G}A)$ with the following property: for every pair of objects A and B in $\mathbf{C}$ and every $f \in \text{Hom}_{\mathbf{C}}(A, B)$ we have $\mathcal{G}(f)\eta_A = \eta_B\mathcal{F}(f)$, i.e., the following diagram commutes:

$$\begin{array}{ccc} \mathcal{F}A & \xrightarrow{\eta_A} & \mathcal{G}A \\ \mathcal{F}(f) \downarrow & & \downarrow \mathcal{G}(f) \\ \mathcal{F}B & \xrightarrow{\eta_B} & \mathcal{G}B \end{array}$$

If each η_A is an isomorphism, η is called a *natural isomorphism* of functors.

Consider the special case where $\mathbf{C} = \mathbf{D}$ and $\mathbf{C}$ is a subcategory of **Set**, and where $\mathcal{F}$ is the identity functor. There is a natural transformation η from the identity functor to $\mathcal{G}$ if whenever $\mathcal{G}$ maps the object A to the object $\mathcal{G}A$ there is a morphism η_A from A to $\mathcal{G}A$, and whenever there is a morphism f from A to B the morphism $\mathcal{G}(f)$ is compatible with f as a map from $\mathcal{G}A$ to $\mathcal{G}B$. In fact $\mathcal{G}(f)$ is uniquely determined by f as a map from the subset $\eta_A(A)$ in $\mathcal{G}A$ to the subset $\eta_B(B)$ of $\mathcal{G}B$. If η is a natural isomorphism, then the value of $\mathcal{G}$ on every morphism is completely determined by η , namely $\mathcal{G}(f) = \eta_B f \eta_A^{-1}$. In this case the functor $\mathcal{G}$ is entirely specified by η . We shall see that some of the examples of functors in the preceding section arise this way.

Examples

- (1) For any categories $\mathbf{C}$ and $\mathbf{D}$ and any functor $\mathcal{F}$ from $\mathbf{C}$ to $\mathbf{D}$ the identity is a natural isomorphism from $\mathcal{F}$ to itself: $\eta_A = 1_{\mathcal{F}A}$ for every object A in $\mathbf{C}$.

- (2) Let R be a ring and let $\mathcal{F}$ be any functor from $R\text{-Mod}$ to itself. The zero map is a natural transformation from $\mathcal{F}$ to itself: $\eta_A = 0_A$ for every R -module A , where 0_A is the zero map from A to itself. This is not a natural isomorphism.
- (3) Let $\mathcal{F}$ be the identity functor from $\mathbf{Grp}$ to itself, and let $\mathcal{G}$ be the abelianizing functor (Example 3) considered here as a map from $\mathbf{Grp}$ to itself. For each group G let $\eta_G : G \rightarrow G/G'$ be the usual projection map onto the quotient group. Then η is a natural transformation (but not an isomorphism) with respect to these two functors. (We call the maps η_G the *natural projection maps*.)
- (4) Let $\mathcal{G} = \mathcal{D}^2$ be the double dual functor from the category of finite dimensional vector spaces over a field K to itself (Example 7). Then there is a natural isomorphism η from the identity functor to $\mathcal{G}$ given by

$$\eta_V : V \rightarrow V^{**} \quad \text{by} \quad \eta_V(v) = E_v$$

where E_v is “evaluation at v ” for every $v \in V$.

- (5) Let $\mathcal{GL}_n$ be the functor from $\mathbf{CRing}$ to $\mathbf{Grp}$ defined as follows. Each object (commutative ring) R is mapped by $\mathcal{GL}_n$ to the group $GL_n(R)$ of $n \times n$ invertible matrices with entries from R . For each ring homomorphism $f : R \rightarrow S$ let $\mathcal{GL}_n(f)$ be the map of matrices that applies f to each matrix entry. Since f sends 1 to 1 it follows that $\mathcal{GL}_n(f)$ sends invertible matrices to invertible matrices (cf. Exercise 4 in Section 1). Let $\mathcal{G}$ be the functor from $\mathbf{CRing}$ to $\mathbf{Grp}$ that maps each ring R to its group of units $R^\times$, and each ring homomorphism f to its restriction to the groups of units (also denoted by f). The *determinant* is a natural transformation from $\mathcal{GL}_n$ to $\mathcal{G}$ because the determinant is defined by the same polynomial for all rings so that the following diagram commutes:

$$\begin{array}{ccc} GL_n(R) & \xrightarrow{\det} & R^\times \\ \mathcal{GL}_n(f) \downarrow & & \downarrow f \\ GL_n(S) & \xrightarrow{\det} & S^\times \end{array}$$

Let $\mathbf{C}$, $\mathbf{D}$ and $\mathbf{E}$ be categories, let $\mathcal{F}$ be a functor from $\mathbf{C}$ to $\mathbf{D}$, and let $\mathcal{G}$ be a functor from $\mathbf{D}$ to $\mathbf{E}$. There is an obvious notion of the composition of functors $\mathcal{GF}$ from $\mathbf{C}$ to $\mathbf{E}$. When $\mathbf{E} = \mathbf{C}$ the composition $\mathcal{GF}$ maps $\mathbf{C}$ to itself and $\mathcal{FG}$ maps $\mathbf{D}$ to itself. We say $\mathbf{C}$ and $\mathbf{D}$ are *isomorphic* if for some $\mathcal{F}$ and $\mathcal{G}$ we have $\mathcal{GF}$ is the identity functor $\mathcal{I}_C$, and $\mathcal{FG} = \mathcal{I}_D$. By the discussion in Section 10.1 the categories $\mathbb{Z}\text{-Mod}$ and $\mathbf{Ab}$ are isomorphic. It also follows from observations in Section 10.1 that the categories of elementary abelian p -groups and vector spaces over $\mathbb{F}_p$ are isomorphic. In practice we tend to identify such isomorphic categories. The following generalization of isomorphism between categories gives a broader and more useful notion of when two categories are “similar.”

Definition. Categories $\mathbf{C}$ and $\mathbf{D}$ are said to be *equivalent* if there are functors $\mathcal{F}$ from $\mathbf{C}$ to $\mathbf{D}$ and $\mathcal{G}$ from $\mathbf{D}$ to $\mathbf{C}$ such that the functor $\mathcal{GF}$ is naturally isomorphic to $\mathcal{I}_C$ (the identity functor of $\mathbf{C}$) and $\mathcal{FG}$ is naturally isomorphic to the identity functor $\mathcal{I}_D$.

It is an exercise that equivalence of categories is reflexive, symmetric and transitive. The example of Affine k -algebras in Section 15.5 is an equivalence of categories (where one needs to modify the direction of the arrows in the definition of a natural

transformation to accommodate the contravariant functors in this example). Another example (which requires some proving) is that for R a commutative ring with 1 the categories of left modules $R\text{-Mod}$ and $M_{n \times n}(R)\text{-Mod}$ are equivalent.

Finally, we introduce the concepts of universal arrows and universal objects.

Definition.

- (1) Let $\mathbf{C}$ and $\mathbf{D}$ be categories, let $\mathcal{F}$ be a functor from $\mathbf{C}$ to $\mathbf{D}$, and let X be an object in $\mathbf{D}$. A *universal arrow* from X to $\mathcal{F}$ is a pair $(U(X), \iota)$, where $U(X)$ is an object in $\mathbf{C}$ and $\iota : X \rightarrow \mathcal{F}U(X)$ is a morphism in $\mathbf{D}$ satisfying the following property: for any object A in $\mathbf{C}$ if φ is any morphism from X to $\mathcal{F}A$ in $\mathbf{D}$, then there exists a unique morphism $\Phi : U(X) \rightarrow A$ in $\mathbf{C}$ such that $\mathcal{F}(\Phi)\iota = \varphi$, i.e., the following diagram commutes:

$$\begin{array}{ccc} X & \xrightarrow{\iota} & \mathcal{F}U(X) \\ & \searrow \varphi & \downarrow \mathcal{F}(\Phi) \\ & & \mathcal{F}A \end{array}$$

- (2) Let $\mathbf{C}$ be a category and let $\mathcal{F}$ be a functor from $\mathbf{C}$ to the category **Set** of all sets. A *universal element* of the functor $\mathcal{F}$ is a pair (U, ι) , where U is an object in $\mathbf{C}$ and ι is an element of the set $\mathcal{F}U$ satisfying the following property: for any object A in $\mathbf{C}$ and any element g in the set $\mathcal{F}A$ there is a unique morphism $\varphi : U \rightarrow A$ in $\mathbf{C}$ such that $\mathcal{F}(\varphi)(\iota) = g$.

Examples

- (1) (*Universal Arrow: Free Objects*) Let R be a ring with 1. We translate into the language of universal arrows the statement that if $U(X)$ is the free R -module on a set X then any set map from X to an R -module A extends uniquely by R -linearity to an R -module homomorphism from $U(X)$ to A (cf. Theorem 6, Section 10.3): Let $\mathcal{F}$ be the forgetful functor from $R\text{-Mod}$ to **Set**, so that $\mathcal{F}$ maps an R -module A to the set A , i.e., $A = \mathcal{F}A$ as sets. Let X be any set (i.e., an object in **Set**), let $U(X)$ be the free R -module with basis X , and let $\iota : X \rightarrow \mathcal{F}U(X)$ be the set map which sends each $b \in X$ to the basis element b in $U(X)$. Then the universal property of free R -modules is precisely the result that $(U(X), \iota)$ is a universal arrow from X to the forgetful functor $\mathcal{F}$.

Similarly, free groups, vector spaces (which are free modules over a field), polynomial algebras (which are free R -algebras) and the like are all instances of universal arrows.

- (2) (*Universal Arrow: Fields of Fractions*) Let $\mathcal{F}$ be the forgetful functor from the category of fields to the category of integral domains, where the morphisms in both categories are *injective* ring homomorphisms. For any integral domain X let $U(X)$ be its field of fractions and let ι be the inclusion of X into $U(X)$. Then $(U(X), \iota)$ is a universal arrow from X to the functor $\mathcal{F}$ (cf. Theorem 15(2) in Section 7.5).
- (3) (*Universal Object: Tensor Products*) This example refers to the construction of the tensor product of two modules in Section 10.4. Let $\mathbf{C} = R\text{-Mod}$ be the category of R -modules over the commutative ring R , and let M and N be R -modules. For each R -module A let $\text{Bilin}(M, N; A)$ denote the set of all R -bilinear functions from $M \times N$ to A . Define a functor from $R\text{-Mod}$ to **Set** on objects by

$$\mathcal{F} : A \longrightarrow \text{Bilin}(M, N; A),$$

and if $\varphi : A \rightarrow B$ is an R -module homomorphism then

$$\mathcal{F}(\varphi)(h) = \varphi \circ h \quad \text{for every } h \in \text{Bilin}(M, N; A).$$

Let $U = M \otimes_R N$ and let ι be the bilinear function

$$\iota : M \times N \rightarrow M \otimes_R N \quad \text{by} \quad \iota(m, n) = m \otimes n,$$

so ι is an element of the set $\text{Bilin}(M, N; M \otimes_R N) = \mathcal{F}U$. Then $(M \otimes_R N, \iota)$ is a universal element of $\mathcal{F}$ because for any R -module A and for any bilinear map $g : M \times N \rightarrow A$ (i.e., any element of $\mathcal{F}A$) there is a unique R -module homomorphism $\varphi : M \otimes_R N \rightarrow A$ such that $g = \varphi \circ \iota = \mathcal{F}(\varphi)(\iota)$.

EXERCISES

1. Let $\mathbf{Nor}\text{-}N$ be the category described in Exercise 1 of Section 1, and let $\mathcal{F}$ be the inclusion functor from $\mathbf{Nor}\text{-}N$ into $\mathbf{Grp}$. Describe a functor $\mathcal{G}$ from $\mathbf{Nor}\text{-}N$ into $\mathbf{Grp}$ such that the transformation η defined by $\eta_G : G \rightarrow G/N$ is a natural transformation from $\mathcal{F}$ to $\mathcal{G}$.
2. Let H and K be groups and let $\mathcal{H} \times$ and $\mathcal{K} \times$ be functors from $\mathbf{Grp}$ to itself described in Exercise 2 of Section 1. Let $\varphi : H \rightarrow K$ be a group homomorphism.
 - (a) Show that the maps $\eta_A : H \times A \rightarrow K \times A$ by $\eta_A(h, a) = (\varphi(h), a)$ determine a natural transformation η from $\mathcal{H} \times$ to $\mathcal{K} \times$.
 - (b) Show that the transformation η is a natural isomorphism if and only if φ is a group isomorphism.
3. Express the universal property of the commutator quotient group — described in Proposition 7(5) of Section 5.4 — as a universal arrow for some functor $\mathcal{F}$.

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